

1. Administrative & Study Identification

Full Study Title: A phase 3, multicenter, randomized, double-blind, placebo-controlled study to evaluate the safety and efficacy of PF-06939926 for the treatment of Duchenne muscular dystrophy **Short Title/Acronym:** CIFFRE0 Study **Protocol Number:** EudraCT 2019-002921-31 **NCT Number:** NCT04281485 **Sponsor Name:** Pfizer Inc. **Investigational Product:** Fordadistrogene movaparvovec (PF-06939926 / rAAV9 Gene Therapy) **Phase of Development:** Phase 3 **Medical Monitor:** Pfizer Clinical Operations Lead / DMD Program Director

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and efficacy of fordadistrogene movaparvovec compared with placebo in male participants with Duchenne Muscular Dystrophy (DMD), specifically assessing the change from baseline in the North Star Ambulatory Assessment (NSAA) at 1 year. **Secondary Objectives:** To assess the 10-meter run/walk velocity, time-to-rise from the floor velocity, mini-dystrophin expression, and long-term safety and tolerability.

2.2 Background & Rationale Duchenne Muscular Dystrophy (DMD) is a severe, progressive disease of the muscles caused by a mutation in the gene responsible for producing dystrophin, leading to muscle degeneration and premature death. Current glucocorticoid treatments aim to delay disease progression but have significant side effects. Fordadistrogene movaparvovec is a recombinant adeno-associated virus serotype 9 (AAV9) gene replacement therapy designed to deliver a shortened, functional "mini-dystrophin" gene to muscle cells. This approach aims to restore functional dystrophin production and improve motor function for patients with DMD.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (Approximately 2/3 of participants assigned to gene therapy and 1/3 to placebo in the first year).

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 99 participants **Number of Sites:** Global Multicenter Trial **Estimated Study Start:** 2020 **Estimated Primary Completion:** 2024 (Note:

Clinical program was discontinued in mid-2024 after failing to meet its primary efficacy endpoint and related safety events).

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male participants between 4 and 8 years of age. 2. Confirmed genetic diagnosis of Duchenne muscular dystrophy (DMD). 3. Ambulatory status. 4. On a stable daily regimen of glucocorticoids.

4.2 Exclusion Criteria 1. Any mutation (exon deletion, exon duplication, insertion, or point mutation) affecting any exon between exon 9 and exon 13, inclusive. 2. A deletion that affects both exon 29 and exon 30. 3. A deletion that affects any exons between 56-71, inclusive.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single, one-time infusion of fordadistrogene movaparvovec (Cohort 1 receives the gene therapy on Day 1; Cohort 2 receives placebo on Day 1 and the gene therapy crossover dose on Day 390). **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** 1 year double-blind efficacy evaluation, followed by an open-label period and long-term safety monitoring for up to 15 years.

5.2 Concomitant & Prohibited Therapy Permitted: Stable daily regimen of glucocorticoids. **Prohibited:** Prior gene therapy for DMD; unapproved adjustments to glucocorticoid maintenance regimens.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Genetic Confirmation, Baseline NSAA
Baseline	Day 0	Randomization, First IV Infusion (PF-06939926 or Placebo)
Interim	Day 390 (Year 1)	1-Year Efficacy Assessment, Crossover Dosing (Placebo cohort receives Gene Therapy)
End of Study	Years 5-15	Long-term safety follow-up, motor function tracking, AE monitoring

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change from baseline in the North Star Ambulatory Assessment (NSAA) at 1 year relative to placebo. **Measurement Method:** Standardized clinical motor function assessment administered by trained evaluators.

7.2 Secondary & Exploratory Endpoints * 10-meter run/walk velocity. * Time-to-rise from the floor velocity. * Mini-dystrophin expression levels (measured via muscle biopsy and quantification of mini-dystrophin-positive fibers).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Assessed during 41 to 43 clinic visits over 5 to 6 years, with ongoing data collection in a 15-year long-term safety follow-up. **Serious Adverse Events (SAEs):** Standard 24-hour reporting. Strict monitoring of AAV-related toxicities is enforced. **Data Monitoring Committee (DMC):** Independent external committee overseeing safety data and pausing/stopping rules globally.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy assessments; Safety Set for all participants receiving the AAV9 infusion. **Sample Size Justification:** \$N=99\$ appropriately powered to detect significant differences in the NSAA score at 1 year between the gene therapy and placebo arms. **Handling of Missing Data:** Standard multiple imputation rules applied for motor function assessments and longitudinal analysis models.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Rigorous onsite visits (up to 43 per participant) and centralized data review. **Audit Trail:** Standard eCRF implementation, with safety and efficacy data periodically locked over the 15-year follow-up period.

11. Study Identifiers & Governance

Primary ID: EudraCT 2019-002921-31 **Registry Number:** NCT04281485
Sponsor/Lead Organization: Pfizer Inc. **Study Title:** CIFFREO
Official Regulatory Title: A phase 3, multicenter, randomized, double-blind, placebo-controlled study to evaluate the safety and efficacy of PF-06939926 for the treatment of Duchenne muscular dystrophy

12. Clinical Framework (DMD Specific)

Primary Condition: Duchenne Muscular Dystrophy (DMD) **Diagnosis Threshold:** Confirmed genetic mutation of the dystrophin gene (excluding specific exon deletions at 9-13, 29-30, and 56-71) **Medical Methodology:** Microdystrophin Gene Replacement Therapy (rAAV9 vector delivery) **Optimization Target:** Restoration of functional mini-dystrophin production in muscle cells

13. Study Procedures & Methodology

Management Pathway: Stable daily glucocorticoids combined with a one-time gene therapy infusion **Education Protocol (HCP):** Clinical training on IV infusion of AAV-based gene therapies and monitoring for immunologic responses **Education Protocol (Patient):** Comprehensive informed consent detailing the mandatory 15-year long-term safety follow-up commitments **Quality Audit Mechanism:** Centralized review of NSAA administration and continuous safety signal tracking

14. Observation & Follow-Up Schedule

Total Duration: 5 to 6 years primary follow-up + 10 years extended long-term safety follow-up (up to 15 years total) **Onsite Visit Cadence:** Approximately 41 to 43 clinic visits over the first 5 to 6 years **Remote Monitoring Frequency:** Routine check-ins as dictated by long-term registry protocols **Checklist Review Interval:** Annual efficacy (NSAA) and safety milestones

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				